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Edge Addition Planarity Suite and Generalized Graph Library

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Summary

A *graph* is a mathematical structure comprising a set V of vertices and a set E of edges, with each edge e corresponding to a pair of vertices called the *endpoints* of e. A graph is *planar* if its vertices can be placed in distinct locations on a plane surface and its edges can be drawn on the plane without the edges intersecting, except at their common vertex endpoints. For a planar graph, a planarity algorithm typically outputs a combinatorial data structure that validation code can use to certify the planarity of the input graph. A planar graph drawing is typically produced by a separate algorithm. On the other hand, if an input graph is not planar, then a planarity algorithm typically outputs a minimal subgraph of the input graph that obstructs planarity. Validation code can use a minimal planarity-obstructing subgraph to certify the non-planarity of an input graph. Moreover, a minimal subgraph obstructing planarity can be used to help decide how to amend an input graph to *planarize* it.

Graphs are used to model a very wide array of real-world problems in which there are objects and relationships between the objects. In artificial intelligence, graphs are used to help with reasoning tasks, such as about the relationship pathways between persons of interest in law enforcement or between genes, tissues, diseases, and medications in bioinformatics research. In Physics, graphs and planarity are used to help compute material phase transitions, particle interactions, and electromagnetic duality. Similarly, in chemistry, graphs may be used to represent atoms and their valence bonds in molecules. Planarity testing can help determine feasible molecular arrangements because the molecular graphs for many compounds of interest must be planar. In electronics, a graph may be used to represent the electrical components and wiring in a circuit, such as transistors and etchings on a silicon wafer. The graph of a circuit must be planar or planarized to eliminate short circuits.

The open source software described in this paper provides a highly performant generalized graph library that also implements state-of-the-art algorithms for planarity-related problems. The generalized graph library and planarity-related algorithms are available to C/C++ and Python graph application developers.

Statement of need

The `Edge Addition Planarity Suite` is an open source software package that was originally developed to implement a new planarity algorithm in (J. M. Boyer & Myrvold, 2004). The Edge Addition Planarity Algorithm is currently regarded as one of two state-of-the-art planarity algorithms due to its conceptual simplicity as well as its speed of execution (contributors, 2026). The software package is available from a GitHub repository (J. M. Boyer, 2026) under a 3-clause BSD license.

To enable development of the Edge Addition Planarity algorithm, as well as the other planarity-related algorithms in (J. M. Boyer, 2006) and (J. M. Boyer, 2012), it was necessary to

develop an extensible, generalized graph library. This generalized graph library provides a high-performance solution for the graph algorithm representation and processing needs of a very wide array of graph applications.

In addition to enabling graph application development in C/C++, the popularity of Python as a scientific application development language has prompted the need for making this graph library and the algorithms it implements available to Python developers. It can be accessed by a Github repository (J. M. Boyer et al., 2026b) and by simply using `pip install` to get `planarity` from the Python package installer (J. M. Boyer et al., 2026a).

State of the field

Several large scientific software packages offer a graph planarity algorithm, which is a complex graph algorithm suitable for benchmarking graph libraries. Mathematica and Wolfram Alpha include planarity algorithms, but these packages are not meant for high performance graph algorithms, so the
Library for Efficient Data Types and and Algorithms (LEDA) is significantly faster because it is an industrial library designed for computational efficiency. Yet, all planarity algorithms implemented in LEDA were shown in (J. M. Boyer et al., 2004) to be several times slower than the Edge Addition Planarity algorithm implementation. LEDA is also has the disadvantage of being C++ only and has a proprietary license. The only graph library shown (in (J. M. Boyer et al., 2004)) to have similar performance to the Edge Addition Planarity Suite is the Public Implementation of a Graph Algorithm Library and Editor (PIGALE). However, PIGALE is also C++ only, has a GPLv2 license, and lacks advanced algorithms for homeomorphic subgraph search. For example, torus embedding algorithms and torus obstruction isolation algorithms are able to operate differently based on the result of a search for a subgraph homeomorphic to *K3, 3* (Myrvold & Kocay, 2011; Myrvold & Woodcock, 2018).

Software design

The generalized graph library in the Edge Addition Planarity Suite has a carefully curated API and an object-oriented architecture. The base **Graph** class structure includes

- base graph, vertex, and edge structure definitions and getter/setter methods;
- graph, vertex and edge allocation, deallocation, and reset methods;
- methods for iterating through all vertices, edges, and edge adjacency lists;
- graph readers and writers for several formats; and
- a dynamic subclassing/extension mechanisms and virtual function table.

The base Graph is extendable with utility methods for exploring a graph, labelling vertices and edges according to a depth-first search (DFS), and for managing connectivity and biconnectivity information. The vertex and edge structures are extended to enable storage of the information generated by the utility methods.

A Graph structure extended with the DFS-related utilities is further extendable to a **Planarity Graph**, which also further extends the vertex and edge structures. The main additional method provided by a Planarity Graph is `gp_Embed()`, which takes a Planarity Graph as input and outputs either a combinatorial planar embedding or a minimal planarity-obstructing subgraph.

Given a Planarity Graph, a number of subclass extensions are available for planarity-related problems. One extension solves outerplanarity. Another implements the planar graph drawing algorithm in (J. M. Boyer, 2006). Three other extensions provide Graph subclasses that solve the homeomorphic subgraph search algorithms appearing in (J. M. Boyer, 2012). All of these extensions further extend the vertex and edge structures and overload the `gp_Embed()` function.

In addition to being available to C and C++ developers, the APIs of the Edge Addition

Planarity Suite and Generalized Graph Library are also made available via the `planarity` Python package (J. M. Boyer et al., 2026b). Cython is used to preserve high performance while also broadening availability to the Python developer community.

A final aspect of this open source software project is our emphasis on software reliability. We perform validation code on `gp_Embed()` and its five overloads on billions of randomly generated graphs up to 10 million vertices and 30 million edges. The GitHub actions that run on every pull request include sanitizer tests on all graphs on 8 vertices. Finally, the Edge Addition Planarity Suite Testing repository (W. B. K. Boyer & Boyer, 2026) provides our multithreaded Python code for validating `gp_Embed()` and its five overloads on the more than one billion graphs having 11 or fewer vertices, as generated by the geng tool in Nauty and Traces (McKay & Adolfo, 2025).

Research impact statement

The journal publication of the Edge Addition Planarity algorithm, (J. M. Boyer & Myrvold, 2004), currently has over 360 scholarly citations according to Google Scholar. The algorithm has been implemented in several large open source projects, including Boost, Magma, nauty and Traces (McKay & Adolfo, 2025), and the Open Graph Drawing Framework (web archive). The source code from the Edge Addition Planarity Suite repository has been directly included in several large open source projects, including SageMath, Digraphs, and over 80 Linux platforms releases including for Debian, Devuan, Kali, openSUSE, Raspbian, Ubuntu, Alpine, ALT, Fedora, FreeBSD, Gentoo, Manjaro, MSYS2, and Parabola, according to repology/edge-addition-planarity-suite (web archive) and repology/planarity (web archive).

The Python package named `planarity` was originally developed in 2013 by Aric Hagberg as a way to get the planarity testing and drawing algorithms from (J. M. Boyer & Myrvold, 2004) and (J. M. Boyer, 2006) to work with graphs from NetworkX (Hagberg et al., 2008). Recently, Hagberg transferred the `planarity` GitHub repository [PlanarityDevelopers:PlanarityGitHub] to the GitHub Graph Algorithms Organization and the `planarity` Python package (J. M. Boyer et al., 2026a) to the PyPI Graph Algorithms Organization, which are both maintained by the authors. On August 1, 2026, the Top PyPI Packages website indicated that the `planarity` Python package had over 700,000 downloads for the preceding month and was ranked in the top 1% of PyPI package downloads (ranked 5262 out of 864,036 projects). The `planarity` Python package has now been included as a standard Python package in several versions of Linux including Debian, Devuan, Kali, Raspbian, and Ubuntu, according to repology/python:planarity (web archive).

AI usage disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, nor the preparation of supporting materials.

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